Leaving a blocked twist tower: 3-isogeny walks and the class ceiling

2026-10-14. Frozen protocol `PROTOCOL.md` (sha256 `86a67503d4b5c4c5…`). Native instrument `examples/twist_walk_blocked_towers.rs`. Both positive controls passed before any data. Scope: 5 catalog curves, 13 towers, depth 4, degree 3 only.

Catalog result (exact computation)

Twist of each of the 13 blocked towers; depth counts are 3-isogeny steps, Frobenius free. `m` is the MMT magic number of the start; `m'` is the least magic number among unblocked nodes within depth 4.

Curve	N	l	n	m	nodes d1,d2,d3,d4	verdict	m’
icv1-f2m1…d2c04152	18	1	18	13	0, 0, 0, 0	no rational 3-isogeny	—
icv1-f2m1…d2c04152	18	3	6	5	0, 0, 0, 0	no rational 3-isogeny	—
icv1-f2m2…54539071	24	1	24	23	0, 0, 0, 0	no rational 3-isogeny	—
icv1-f2m2…2d0c0e8c	24	1	24	22	1, 0, 0, 0	no unblocked twist within 4 steps	—
icv1-f2m2…2d0c0e8c	24	2	12	11	1, 0, 0, 0	unblocked at depth 1	12
c2pnb208w1	208	1	208	14	0, 0, 0, 0	no rational 3-isogeny	—
c2pnb208w1	208	2	104	7	0, 0, 0, 0	no rational 3-isogeny	—
c2pnb208w1	208	13	16	14	0, 0, 0, 0	no rational 3-isogeny	—
c2pnb208w1	208	26	8	7	0, 0, 0, 0	no rational 3-isogeny	—
c2pnb368w1	368	1	368	13	1, 3, 0, 0	no unblocked twist within 4 steps	—
c2pnb368w1	368	2	184	7	1, 3, 0, 0	unblocked at depth 1	8
c2pnb368w1	368	23	16	13	1, 3, 0, 0	no unblocked twist within 4 steps	—
c2pnb368w1	368	46	8	7	1, 3, 0, 0	unblocked at depth 1	8

Of 13 towers, 3 are unblocked after one step (`N=24, l=2`; `c2pnb368w1, l=2` and `l=46`). 7 have no rational 3-isogeny at all (three curves, including `c2pnb208w1`). 3 have neighbours and stay blocked: the two `l = 1` towers and `c2pnb368w1, l = 23`.

Random blocked starts, for context (not tested against the catalog)

2000 random starts per configuration, same field, `Tr_(K\/k)(b) = 0` forced; 95% Clopper–Pearson intervals. The last column counts escapes whose least `m'` equals `n` (the generic maximum).

N	l	n	starts	has a 3-isogeny	escaped within 4 steps	escapes with m’ = n
18	1	18	2000	74.2% (0.722–0.761)	0.0% (0.0000–0.0018)	0/1
18	3	6	2000	75.4% (0.735–0.773)	64.6% (0.6246–0.6670)	1225/1292
24	1	24	2000	74.3% (0.723–0.762)	0.0% (0.0000–0.0018)	0/1
24	2	12	2000	73.7% (0.717–0.756)	36.8% (0.3468–0.3896)	630/736
208	1	208	2000	75.2% (0.732–0.770)	0.0% (0.0000–0.0018)	0/1
208	2	104	2000	74.3% (0.723–0.762)	36.0% (0.3394–0.3820)	721/721
208	13	16	2000	75.0% (0.731–0.769)	75.0% (0.7309–0.7693)	1501/1501
208	26	8	2000	74.7% (0.727–0.766)	74.7% (0.7273–0.7659)	1494/1494
368	1	368	2000	75.6% (0.737–0.775)	0.0% (0.0000–0.0018)	0/1
368	2	184	2000	77.1% (0.752–0.789)	39.9% (0.3775–0.4208)	798/798
368	23	16	2000	74.8% (0.728–0.767)	74.8% (0.7284–0.7669)	1496/1496
368	46	8	2000	74.4% (0.724–0.763)	74.4% (0.7243–0.7630)	1488/1488

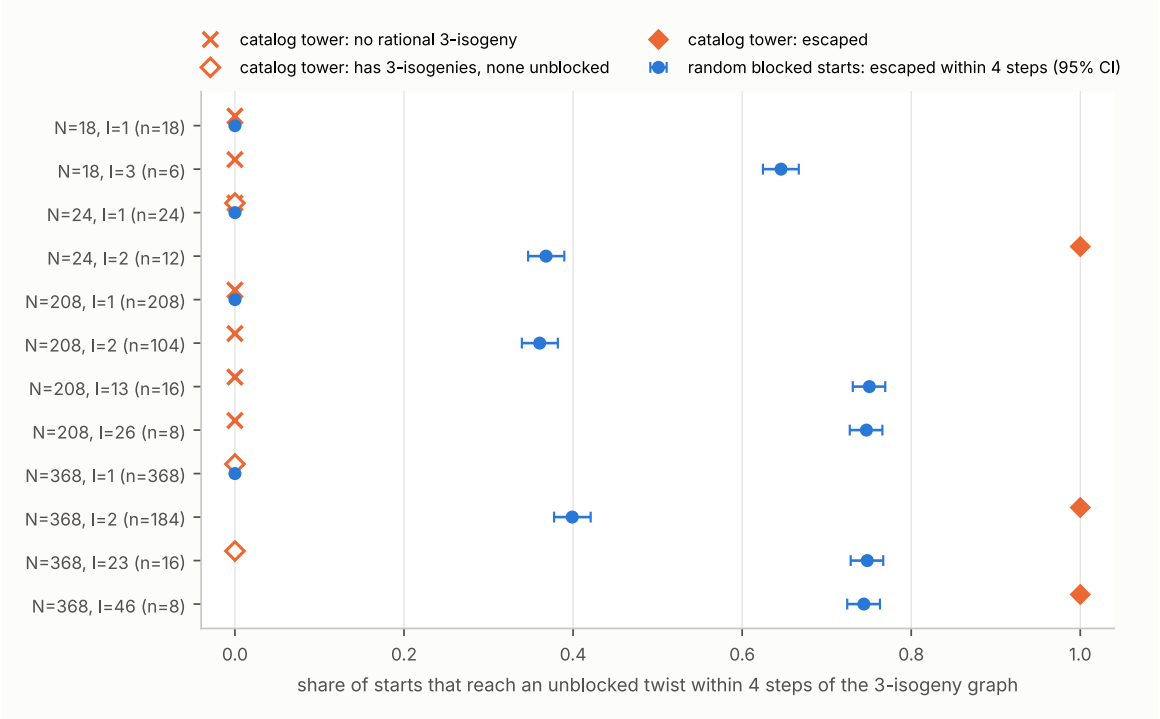


Figure 1: Escape rate of random blocked starts within 4 steps (blue, 95% interval) and the catalog towers (orange). Source: `summary.csv`.

Class ceiling (exact computation, finite)

The twist-side isogeny class of `b` is `{b' : t(b') = t(b)}`. For every field `N = 8…22` and every `l | N` with `n = N/l` even we counted the trace classes holding a blocked curve and those that also hold an unblocked one (`class_ceiling.csv`). For `l = 1` no class is mixed in any of the 8 field sizes (0 mixed classes in total), so no isogeny of any degree leaves an `l = 1` blocked tower in those fields. For `l >= 2`, 60% to 99.9% of blocked curves lie in a mixed class, so an escape exists inside the class.

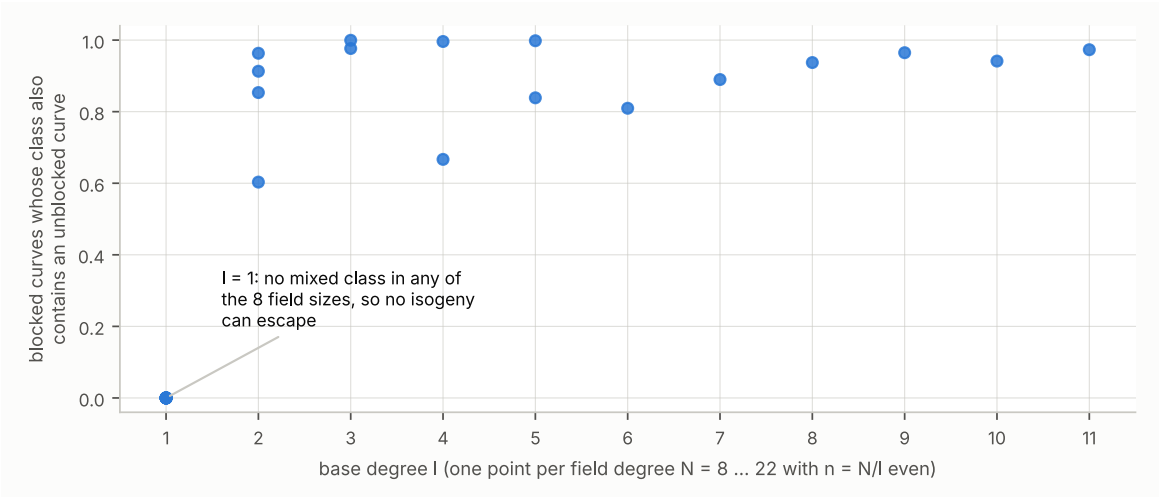


Figure 2: Share of blocked curves whose isogeny class also contains an unblocked curve, by `l`. Source: `class_ceiling.csv`.

Claims and scope

- **Exact computation:** the 13 catalog outcomes above; the class counts for `N = 8…22`.
- **Observed on a sample:** 0 of 2000 random starts escape an `l = 1` tower in each of four fields (upper limit 0.0018), as the class counts require; 36–40% escape at `l = 2`, 65% at `N = 18, l = 3`, and 74–75% at `l >= 13`.
- **Not tested:** that the catalog differs from random starts. Five independent graphs cannot support that test at the corrected level (`PROTOCOL.md`). One catalog escape is far from generic: `c2pnb368w1` at `l = 2` reaches `m' = 8` where random escapes all have `m' = n = 184`; the other two escapes have `m' = n`. That contrast is descriptive.
- **Conjecture, not proved here:** `Tr_(K\/F_2)(b)` is constant on each trace class for all even `N`; the exhaustive counts support it for `N <= 22`.
- **Not shown:** that any twist is attackable. An unblocked twist tower removes the Lemma 6 obstruction only; the twist's group order and cofactor, the cover's genus and the descent cost are separate. Degrees other than 3 are not walked, and depth is capped at 4.